Algorithm 1 Meta-evolution

Inputs: baseline; G generations; O offspring per generation; P population size; tasks T; N_{init} initial seeds; B reevaluation seeds per generation; N_{val} validation seeds; reevaluation `strategy` $\in \{\texttt{TTTS}, \texttt{uniform}\}$.

`population` $\leftarrow$ [`baseline`]
`archive` $\leftarrow \{\}$
for $g = 1, \ldots, G$ **do**
  for $o = 1, \ldots, O$ **do** ▷ *Sample offspring*
    `parents` $\sim$ SAMPLEPARENTS(`population`)
    `prompt` $\sim$ PROMPT($T, g,$ `parents`)
    $x_o \sim$ LLM(`prompt`)
    EVALUATE(x_o, T, N_{init})
    `archive` $\leftarrow$ `archive` $\cup \{x_o\}$
  $\hat{\mu}_x, n_x \leftarrow$ mean score, seed count of $x \in$ `archive` ▷ *Reevaluate*
  if `strategy` $=$ `TTTS` **then**
    $\psi_x \leftarrow$ top-two Thompson probability of x under $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$
    Draw B members from `archive` with probability $\propto \psi_x$
  else if `strategy` $=$ `uniform` **then**
    Split B evenly over the top-P members of `archive` by $\hat{\mu}_x$
  Evaluate each drawn member on new seeds and update $\hat{\mu}_x, n_x$
  `population` $\leftarrow \{\arg\max_{x \in \texttt{archive}} \hat{\mu}_x(t) \mid t \in T\}$ ▷ *MAP-Elites, one niche per task*
  Fill remaining slots by highest overall $\hat{\mu}_x$
  $x^*_g \leftarrow \arg\max_{x \in \texttt{population}} \hat{\mu}_x$ ▷ *Validate generation's best*
  `val`$[x^*_g] \leftarrow$ EVALUATE($x^*_g, T_{\text{val}}, N_{\text{val}}$)
return $\arg\max_x$ `val`$[x]$